LABEL AUTOMATOR

Exploration of 3 different approaches
(scripting, coding, and AI prompting)

The problem to solve

- Easy to forget specific data required by complex labelling regulations.

- Errors and slowness of manually importing data.

Údarás Sábháilteachta Bia na hÉireann

Food Safety Authority of Ireland

The problem to solve

Údarás Sábháilteachta Bia na hÉireann
Food Safety Authority of Ireland

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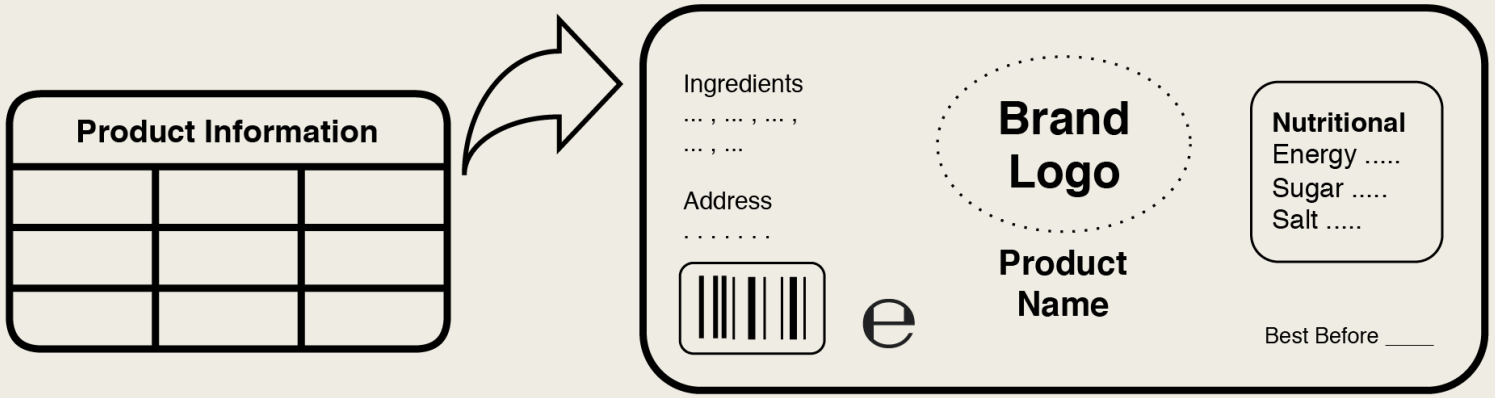
Food Alerts

17 APR	Recall of specific batches of Crowes Farm Black & White Pudding Roulade
10 APR	Recall of a batch of Chef Select Carbonara Pasta Bake
9 APR	Recall of specific batches of Tesco Finest Summer Edition Chocolate Affogato Dessert
9 APR	Recall of a batch of Pettitt's Cook at Home Basil & Pesto Chicken Fillets

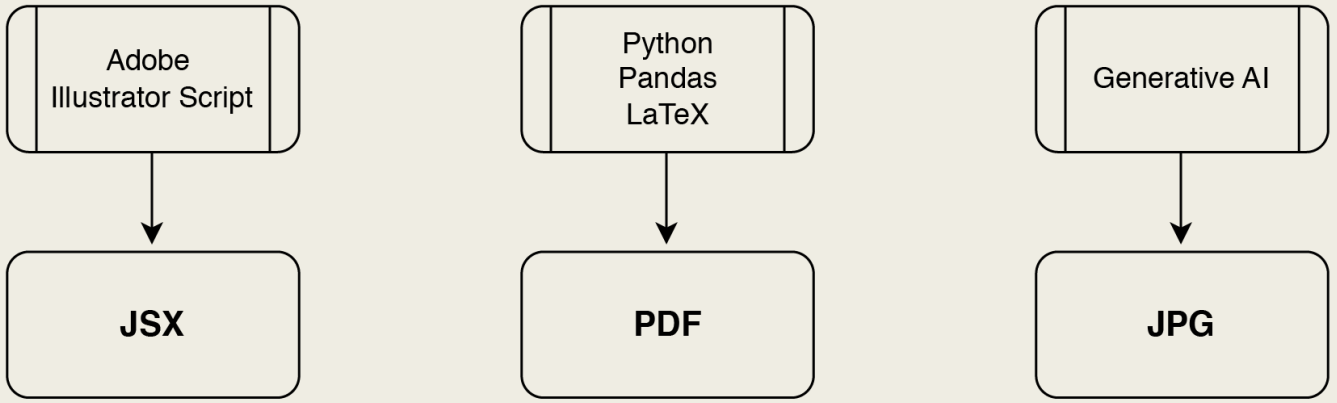
Allergen Alerts

18 APR	Possible presence of gluten in a batch of M&S Collection Authentic Greek Yogurt with Vanilla
20 MAR	Incorrectly declared sulphur dioxide in a batch of Savana Dried Apricots
24 FEB	Incorrectly declared allergens in Ketoclassic 3:1 Porridge, Muesli and Savoury
19 FEB	Undeclared almonds in a batch of mispacked Cottage Delight Garlic Stuffed Gordal Olives

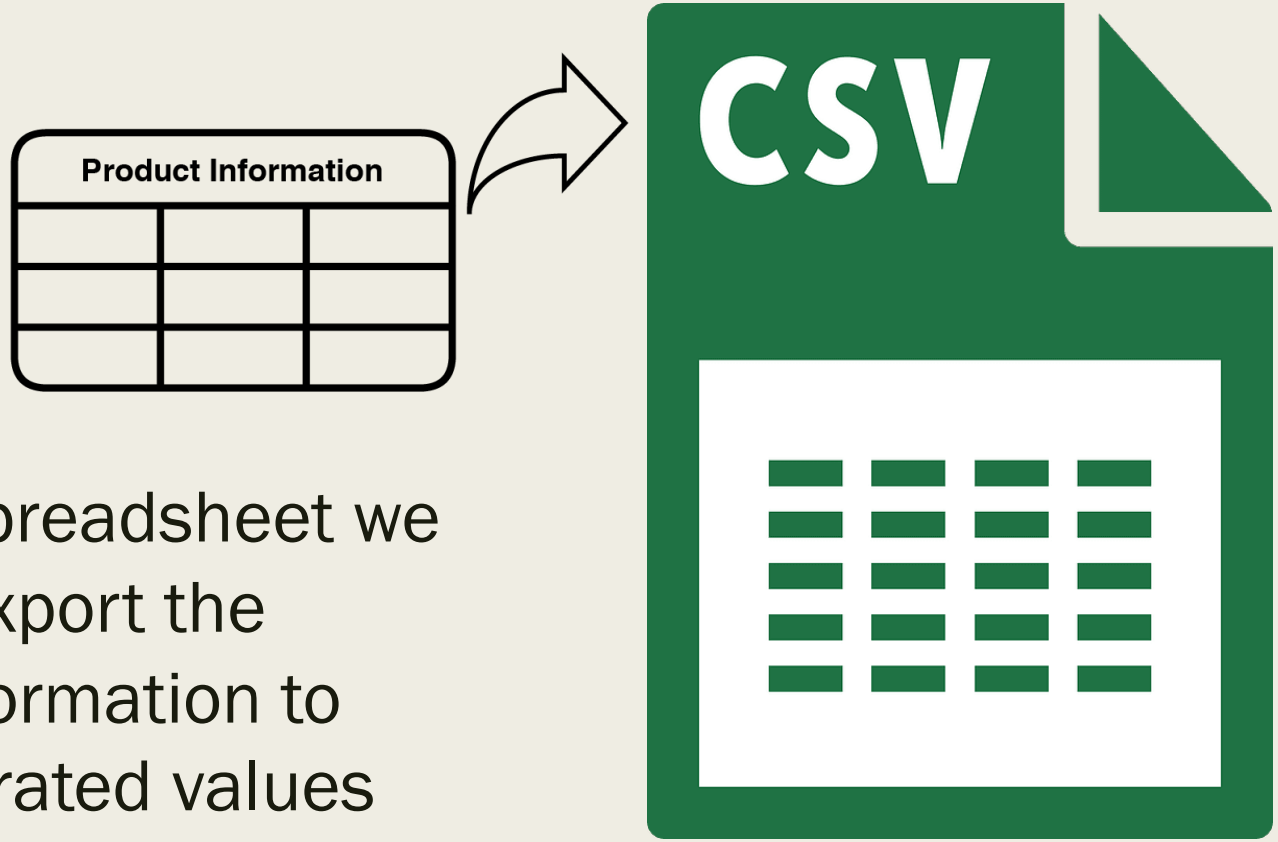
Label Design & Compliance Workflows



Workflows



Export data



- From the spreadsheet we manually export the product information to coma separated values

1. ADOBE ILLUSTRATOR SCRIPT

Script to import product data into the working document

How the Illustrator script works

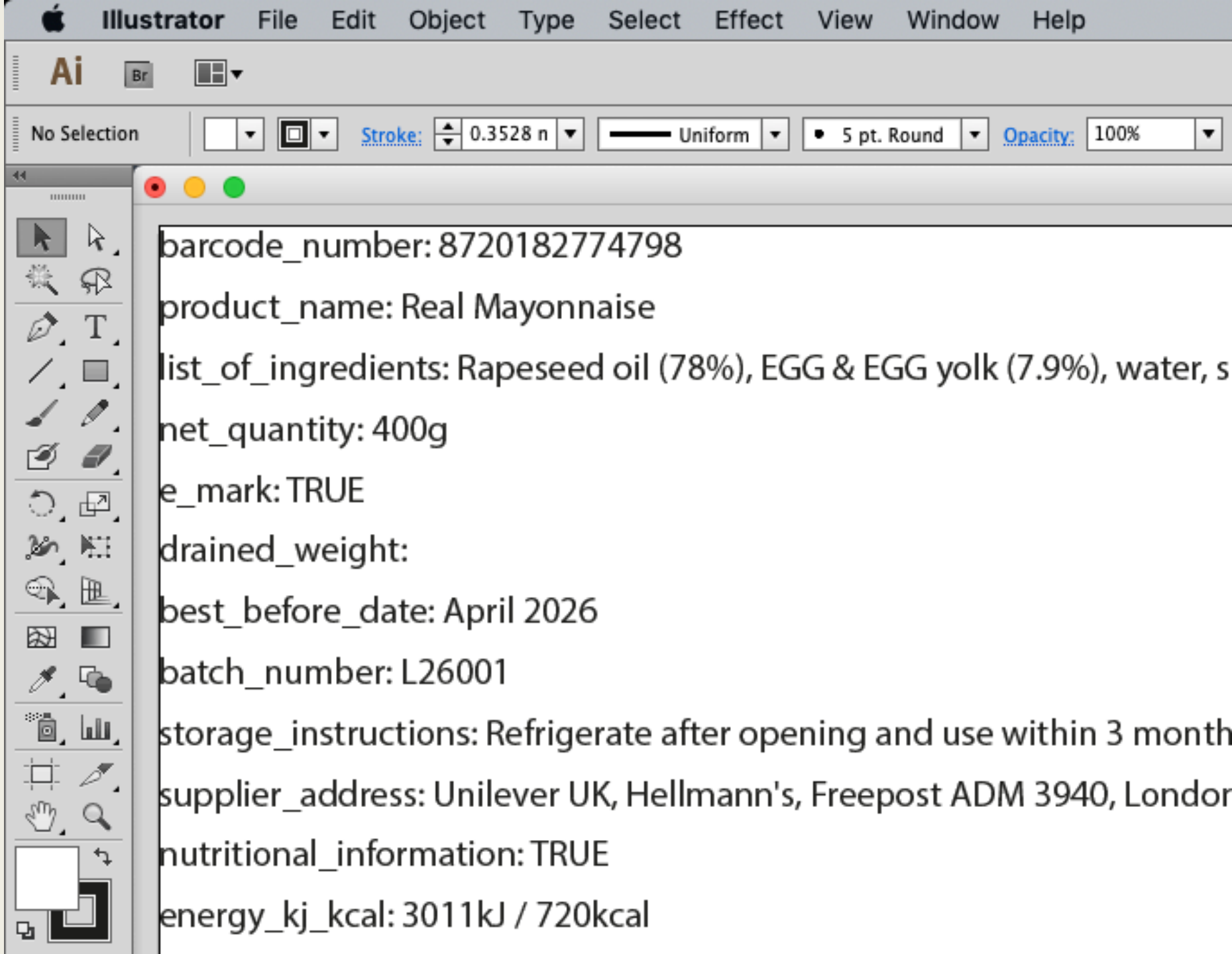
JavaScript JS
ECMAScript 3(1999)

- From Adobe Illustrator we will select a JavaScript to import coma separated values. **JSX**

- Uses Adobe's specific functions to insert and format the information. **ExtendScript.**

Ai Adobe Illustrator

Result

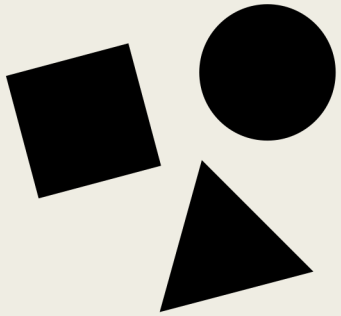

Illustrator File Edit Object Type Select Effect View Window Help
Ai
No Selection
Stroke: 0.3528 n
Uniform
5 pt. Round
Opacity: 100%
barcode_number: 8720182774798
product_name: Real Mayonnaise
list_of_ingredients: Rapeseed oil (78%), EGG & EGG yolk (7.9%), water, s
net_quantity: 400g
e_mark: TRUE
drained_weight:
best_before_date: April 2026
batch_number: L26001
storage_instructions: Refrigerate after opening and use within 3 month
supplier_address: Unilever UK, Hellmann's, Freepost ADM 3940, Londor
nutritional_information: TRUE
energy_kj_kcal: 3011kJ / 720kcal

What the Illustrator script is good for?

- Time saving; one click import
- Avoids copy-paste mistakes
- Adapts to table changes

Limitations

Basic functionality
in this case



Adobe only

Adobe

2. CODING

Python code to create the final label in PDF

How the coding approach works

- Template crafted in LaTeX typesetting.

How the coding approach works

- Uses Python Pandas library to read a CSV file.

How the coding approach works

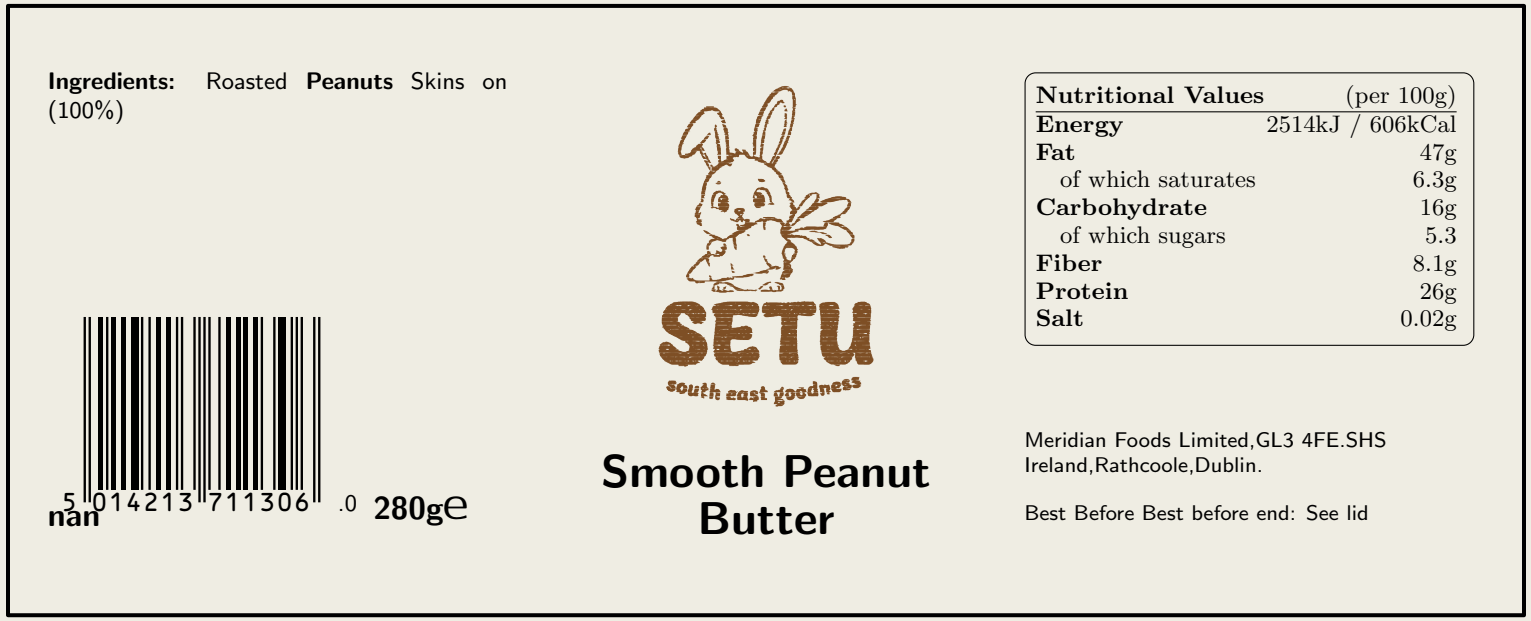
- Python code populates a LaTeX (La-Tek) template with the read fields.

How the coding approach works

- Generates a PDF from the template using **pdflatex.**
- Saves label (TeX and PDF) to respective folder.

LaTeX

Result of the coding approach

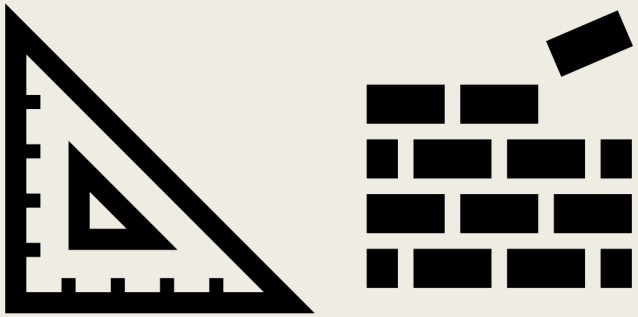


What the coding approach is good for?

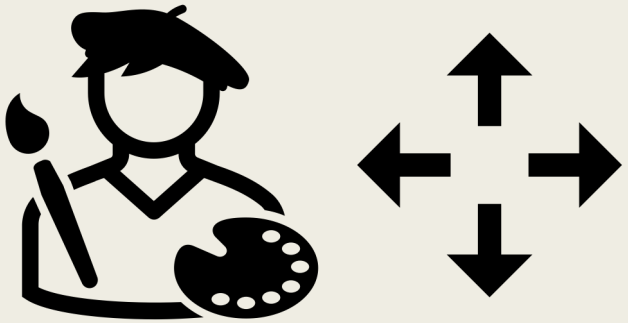
- Generating many similar labels
- No need for Adobe software
- LaTeX package supports typesetting in multiple languages

Limitations of the coding approach

Complex to craft a typesetting template

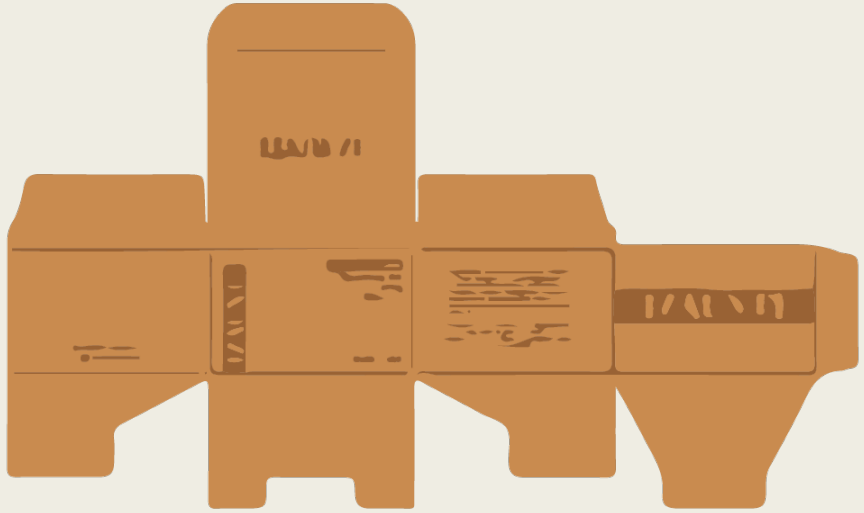


Design adjustments still needed



Other applications of the coding approach

Outer Case Design



White Label

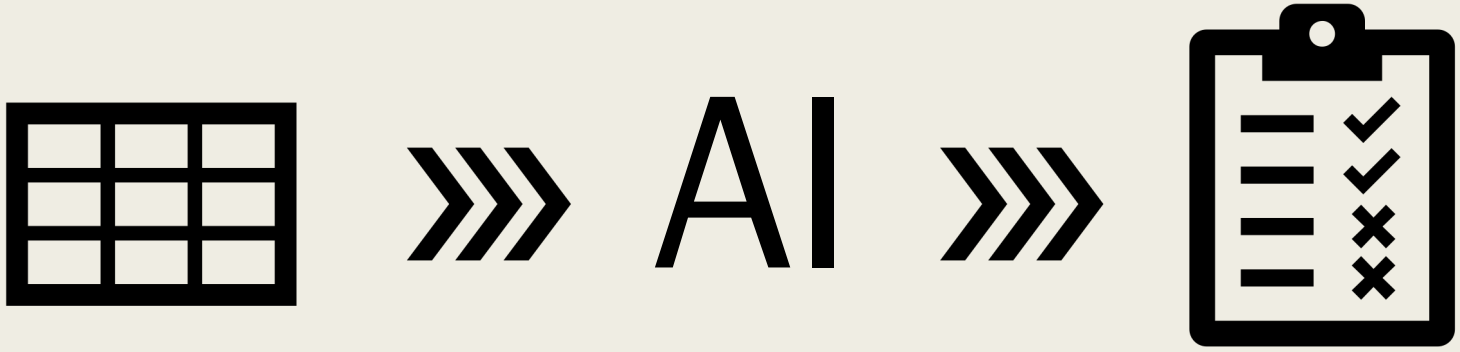
- Replace brand logo, business address and barcode

3. AI PROMPTING

Prompt AI to generate a label

How AI prompting works

- Upload a Coma Separated Value file into AI chat
- Prompt “check for compliance with labelling regulations”



Result of generative AI

Gemini

Result of generative AI

perplexity

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What AI is good for?

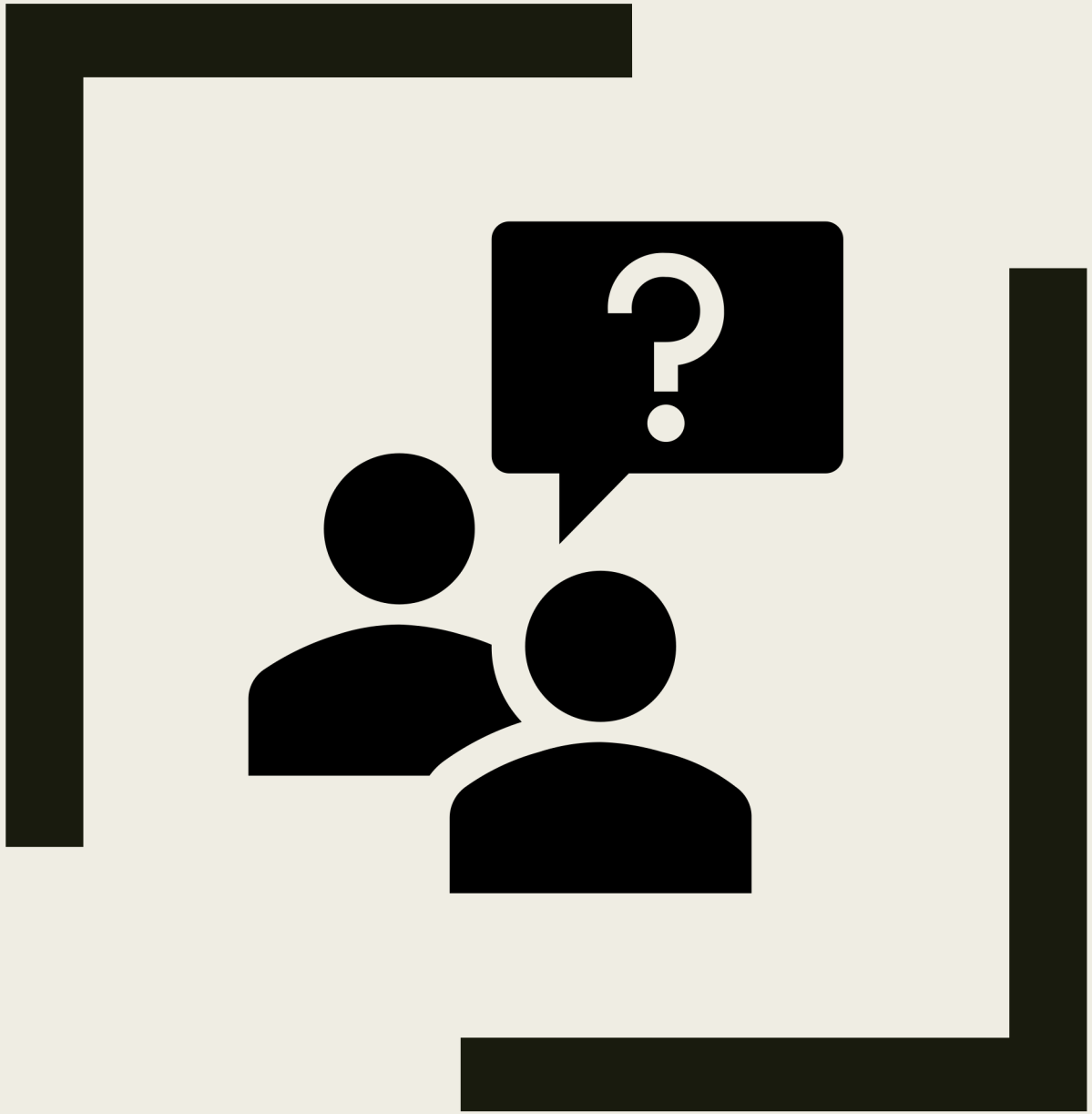
- Compliance check in **text mode.**
- Design brainstorming.
- Label preview and visual changes using natural language

Limitations of AI

Still needs to be reviewed by a human.

Intellectual property infringement.

Generative AI hallucinations.



Questions?

Thank you